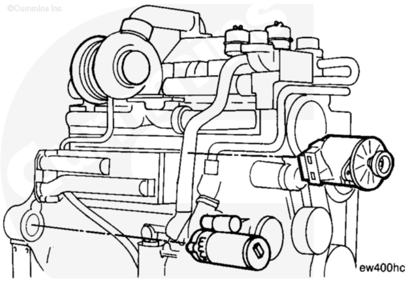
Trainings

Install

Install all accessories and brackets that were removed from the engine.

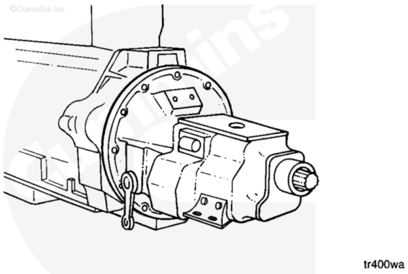


⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : On applications where the rear engine mounts are attached to the driven unit, it is sometimes necessary to install the engine and drive unit as an assembly.

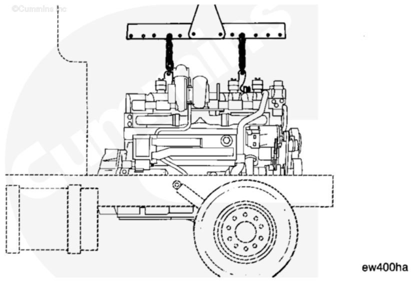
Inspect engine lifting brackets for cracks or other damage. Do **not** attempt to lift engine if cracks or other damage is visible.

Engine weight (wet): 2045 kg [4508 lb]

See the OEM troubleshooting and repair manual for the weight of the drive unit. The engine lifting fixture, Part Number 3162871, is designed to lift a maximum of 2722 kg [6000 lb].

Use a properly rated hoist and engine lifting fixture, Part Number 3162871, to install the engine.

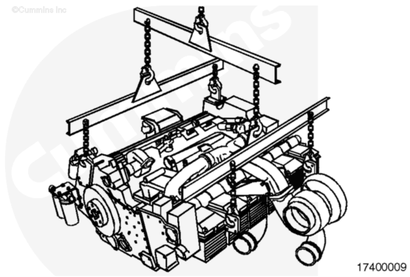
Use 2 Lifting Hooks, Part Number 3193091, to lift the engine. Place the hooks between the cylinder heads, under the lifting lugs.



Rail Applications

⚠ WARNING ⚠

Engine weighs 2045 kg [4508 lb]. Use a properly rated hoist and engine lifting fixture, Part Number 3162871, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.



⚠ WARNING ⚠

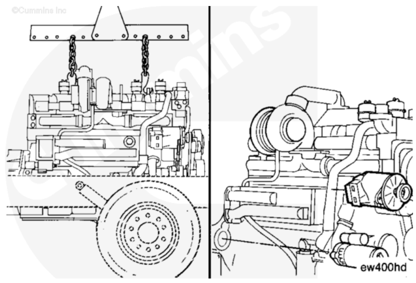
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Inspect engine lifting brackets for cracks or other damage. Do **not** attempt to lift engine if cracks or other damage is visible.

Install three engine lifting fixtures, Part Number 3162871, to engine lifting brackets as shown.

Install the engine.

Align the engine in the chassis and tighten the engine mounting capscrews. See the OEM troubleshooting and repair manual for the torque specifications.

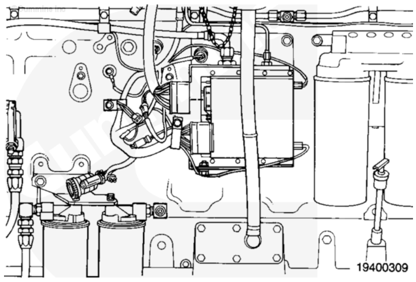


Connect all engine and chassis mounted accessories that were removed.

Make certain that all lines, hoses, and tubes are in good condition, properly routed, and fastened to prevent damage.

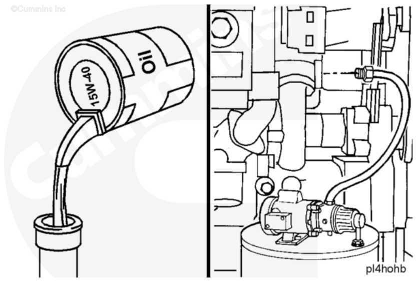
Connect the ECM harness at the junction block at the rear fuel pump side for the engine to the OEM wiring harness. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/13/13-

019-031.html) in the Electronic Control System, QSK19 CM850, Modular Common Rail System Series Engine, Bulletin 4021493.



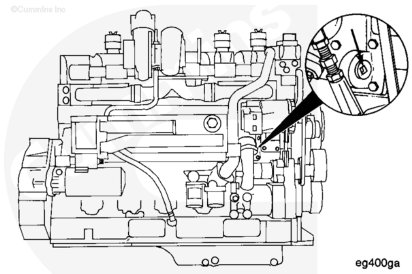
Fill the engine with clean 15W-40 lubricating oil.

Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html) for total oil system capacity.



⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after rebuild to reduce the possibility of internal component damage. Do not prime the system from the bypass filter as the filter will be damaged.

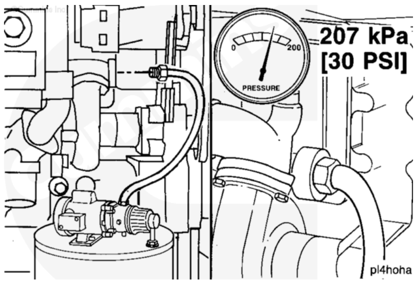


Remove the large plug [1-1/16 inch X 12 UNF] from the oil cooler housing.

Use a pump capable of supplying 205 kPa [30 psi] continuous pressure. Connect the pump to the front of the engine oil cooler as shown.

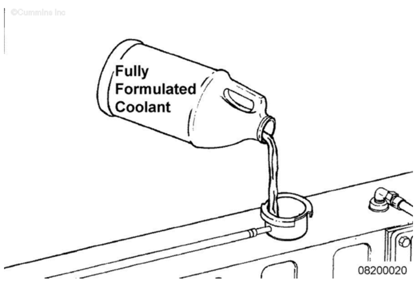
Use a supply of clean oil. Turn the pump to the ON position. Check the engine oil pressure gauge. When the gauge indicates oil pressure, begin monitoring the oil level in the oil pan.

Stop the pump when the oil level reaches the low mark on the dipstick.



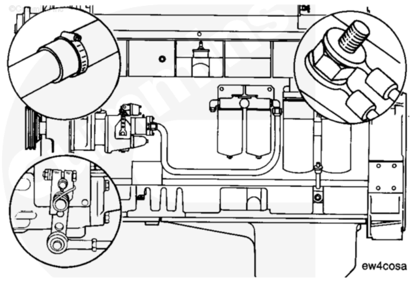
Fill the cooling system with fully formulated coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html).

Coolant capacity (engine **only**): 30 liter [32 qt]



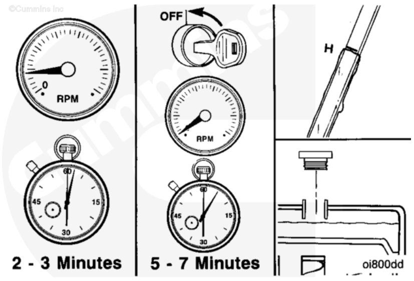
See the equipment manufacturer's specifications for radiator and cooling system capacity.

Complete a final inspection to be certain that all hoses, wires, linkages, and components have been properly installed and tightened.

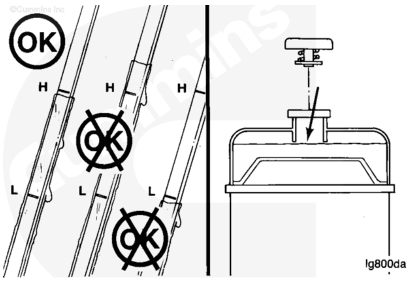


Operate the engine at low idle, for 2 to 3 minutes, with the radiator cap off until the system is vented of air.

Stop the engine and wait 5 to 7 minutes for the oil to drain into the oil pan and check the oil and coolant levels again.



Add oil or coolant to obtain the correct level if necessary. Refer to the Procedure 018-018 (/qs3/pubsys2/xml/en/procedures/20/20-018-018.html) for coolant specifications.



Operate the engine for 8-10 minutes to check for proper operation, unusual noises, and coolant, fuel or lubricating oil leaks.

Repair all leaks and component problems.